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3.3P

Drawing.cs

```
using System.ComponentModel.Design;
using System.IO.Pipelines;
using SplashKitSDK;

namespace ShapeDrawer
{
    public class Drawing
    {
        private readonly List<Shape> _shapes;
        private Color _background;

        public Drawing(Color background)
        {
            _shapes = new List<Shape>();
            _background = background;
        }
        public Drawing() :this(Color.White)
        {

        }
        public Color Background
        {
            get {return this._background;}
            set {this._background = value;}
        }
        public int ShapeCount
        {
            get{return this._shapes.Count;}
        }
        public void AddShape(Shape s)
        {
            _shapes.Add(s);
        }
        public void RemoveShape(Shape s)
        {
            _ = _shapes.Remove(s);
        }
        public void Draw()
        {
            SplashKit.ClearScreen(_background);
            foreach (Shape s in _shapes)
            {
                s.Draw();
            }

        }
        public void SelectShapesAt(Point2D pt)
        {
            foreach (Shape s in _shapes)
            {
                s.Selected = s.isAt(pt);
            }
        }
```

```
        public List<Shape> SelectedShapes
        {
            //read-only
            get
            {
                List<Shape> selectedShapes = new List<Shape>();
                foreach(Shape s in _shapes)
                {
                    if(s.Selected) selectedShapes.Add(s);
                }
                return selectedShapes;
            }
        }
    }
}
```

Shape.cs

```
using SplashKitSDK;

namespace ShapeDrawer
{
    public class Shape
    {
        private Color _color;
        private float _x;
        private float _y;
        private int _width;
        private int _height;

        private bool _selected = false;

        public Shape(int param)
        {
            _color = Color.LightGreen;
            _x = 0.0f;
            _y = 0.0f;
            _width = param;
            _height = param;
        }
        public Color color
        {
            get { return _color; }
            set { _color = value; }
        }
        public float x
        {
            get { return _x; }
            set { _x = value; }
        }

        public float X { get; internal set; }

        public float y
        {
            get { return _y; }
            set { _y = value; }
        }
        public int width
        {
```

```
            get { return _width; }
            set { _width = value; }
        }
        public int height
        {
            get { return _height; }
            set { _height = value; }
        }
        public bool Selected
        {
            get {return _selected;}
            set {_selected = value;}
        }
        public void Draw()
        {

            if (this._selected)
            {
                this.DrawOutline();
            }
            SplashKit.FillRectangle(_color, _x,_y, _width, _height);
        }
        public bool isAt(Point2D pt)
        {
            return (pt.X >= _x && pt.X <= _x + _width && pt.Y >= _y && pt.Y <= _y + _height);
        }
        public void DrawOutline()
        {
            int outlineWidth = 7;
            SplashKit.FillRectangle(Color.Black, _x-outlineWidth, _y-outlineWidth,
_width+outlineWidth*2, _height+outlineWidth*2);

        }


    }

}
```

Program.cs

```
using System;
using SplashKitSDK;

namespace ShapeDrawer
{
    public class Program
    {
        public static void Main()
        {
            Window window = new Window("Shape Drawer", 800, 600);
            Drawing myDrawing = new Drawing();
            do
            {
                SplashKit.ProcessEvents();
                SplashKit.ClearScreen();

                if (SplashKit.MouseClicked(MouseButton.LeftButton))
                {
                    Shape myShape = new Shape(100);
```

```
                myShape.x = SplashKit.MouseX();
                myShape.y = SplashKit.MouseY();
                myDrawing.AddShape(myShape);
            }
            if (SplashKit.KeyTyped(KeyCode.SpaceKey))
            {
                myDrawing.Background = SplashKit.RandomColor();
            }
            if (SplashKit.MouseClicked(MouseButton.RightButton))
            {
                Console.WriteLine("Right mouse button clicked");
                myDrawing.SelectShapesAt(SplashKit.MousePosition());
            }
            if (SplashKit.KeyTyped(KeyCode.DeleteKey) ||
SplashKit.KeyTyped(KeyCode.BackspaceKey))
            {
                Console.WriteLine("Delete key pressed");
                foreach(Shape s in myDrawing.SelectedShapes)
                {
                    myDrawing.RemoveShape(s);
                }
            }
            myDrawing.Draw();
            SplashKit.RefreshScreen();
        }while(!window.CloseRequested);

    }

  }
}
```